

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 12

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>1 mark for each point.</i> • Vehicles: high energy density (200 Wh kg^{-1}) minimises battery mass so range is greater; many cycles and high efficiency (1) • Stationary storage: mass is unimportant so the low energy density of lead-acid does not matter; lowest cost per kWh; well-established, recyclable (1) • Lead-acid drawbacks: fewer cycles and lower efficiency; lithium fire risk needs management (1) • Nickel-cadmium largely replaced because of cadmium toxicity and cost (1)	4
01.2	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Recharging speed (1) • Self-discharge during non-use (1) • Availability of raw materials (lithium, cobalt) (1) • Recyclability / disposal (1)	2
01.3	<i>1 mark for each point.</i> • Parked electric vehicles plugged into the grid form a very large distributed battery (1) • At peaks a small proportion of their stored energy is fed back to the grid; surplus grid energy recharges them - avoiding costly, low-efficiency open-cycle gas turbines (1)	2
01.4	<i>1 mark for each point.</i> • Surplus electricity electrolyses water to hydrogen, which can be converted to methane (1) • Methane is fed into the existing natural gas pipe network and storage, so no new distribution system is needed (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 223-224; AQA spec 3.3.4.1.	

Question	Marking guidance	Marks
02.1	<i>1 mark for each point.</i> • 1 nitrogen fixation; 2 nitrification (1) • 3 denitrification; 4 ammonification (1) • 5 ionisation; 6 leaching (3 marks for all six, 2 for four or five, 1 for two or three) (1)	3
02.2	<i>1 mark for each point.</i> • Drainage makes the soil aerobic (1) • Aerobic nitrifying bacteria (and nitrogen fixers) increase so more ammonium is converted to nitrate (1) • Anaerobic denitrifying bacteria decline so less nitrate is lost as nitrogen gas (1)	3
02.3	<i>1 mark for each point.</i> • Nitrogen gas is very unreactive so most organisms cannot use it directly (1) • Only nitrogen fixation (bacteria, lightning, Haber process) converts it to usable ammonium / nitrate; the cycle recycles a small usable amount (1)	2
02.4	<i>1 mark for each point.</i> • Rhizobium bacteria in their root nodules fix atmospheric nitrogen into ammonium / proteins (1) • When the plants die / are ploughed in, decomposition and nitrification release nitrates for the following crop (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 152-154; AQA spec 3.2.4.3.	
03.1	<i>1 mark for each point.</i> • There is an environmental gradient (distance from the hedge / changing conditions) so samples must be taken at regular intervals along it to detect the change (1) • Random sampling might leave gaps / miss the pattern; systematic sampling avoids bias while covering the gradient (1)	2
03.2	<i>1 mark for each point.</i> • Sample at close intervals (eg every 1 m) first and see at what spacing the variations are still detected without unnecessary data; wider intervals (eg 4 m) may miss peaks and troughs (1) • Take increasing numbers of samples at one site and calculate the running mean; the number needed is where the mean stops fluctuating / stabilises (1) • Calculate standard deviation to assess scatter: more variable data need more samples (1) • Test the method (extraction technique, timing) for reliability before the main study (1)	4
03.3	<i>Up to 4 marks in total; award marks in proportion to the completeness of each point.</i> • Remove surface vegetation; pour an irritant solution (dilute mustard / detergent in water) onto the area until saturated (and around it), keep adding as it infiltrates (up to 2) • Collect worms that come to the surface for a standardised time; count / identify / weigh (up to 2) • Limitations: solution may not percolate impermeable soils or reach deep worms; worms may move sideways or downwards; some may die or not respond; slow-moving worms missed (up to 2)	4

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-393; AQA spec 3.7.1.	
04.1	<i>1 mark for each point.</i> - Clay particles are very small and carry electrostatic charges on their surfaces so they repel each other and stay suspended (1) - Flocculants (aluminium sulfate / alum or polyelectrolytes) are added and mixed to neutralise the charges (1) - The particles clump together (flocs) and settle in the clarifier tank (1)	3
04.2	<i>1 mark for each point.</i> - Increases dissolved oxygen and removes hydrogen sulfide (bad-egg smell) from anaerobic sources (1) - Oxidises some dissolved metals (eg iron) so they become insoluble and can be removed; improves taste (1)	2
04.3	<i>1 mark for each point.</i> - Peaty water contains organic matter which reacts with chlorine to produce toxic substances (eg trihalomethanes) (1) - Ozone / UV kill pathogens without forming these by-products (though chlorine / chloramine keeps water sterile during distribution) (1)	2
04.4	<i>1 mark for each point.</i> - Ion exchange: toxic ions adsorb onto polymer resin beads or zeolite (also used to soften water by removing calcium / magnesium) (1) - Reverse osmosis / distillation (desalination) - expensive, energy-intensive (1) - Activated carbon: organic molecules causing odour / taste and pesticides adsorb onto the high-surface-area carbon particles (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 126-128; AQA spec 3.2.2.4.	
05.1	<i>1 mark.</i> The length of time a pollutant remains in the environment before it is broken down / degraded (chemically or biologically) (1)	1
05.2	<i>Any 4 from the points below - 1 mark each, maximum 4.</i> - Light: UV light degrades many pesticides (eg neonicotinoids break down fastest in sunlight) so persistence is shorter in bright conditions (1) - Temperature: higher temperatures increase the rate of chemical breakdown / microbial decomposition (1) - Soil microbes / oxygen: aerobic microbial activity breaks pesticides down so persistence is shorter in well-aerated, biologically active soils (1) - pH / presence of other chemicals may increase or decrease the rate of degradation (1) - Adsorption onto clay / organic matter may protect from degradation and increase persistence (1)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 263-266; AQA spec 3.4.3.2.8.	
06.1	<i>1 mark.</i> Temperature inversion (1)	1

Question	Marking guidance	Marks
06.2	<i>1 mark for each point.</i> • On night A, temperature falls with altitude; warm effluent gases are less dense / more buoyant than the surrounding air so they rise (1) • As they rise they cool but the surrounding air is also cooler so they stay buoyant and continue to rise, dispersing and diluting (1) • On night B the air near the ground is colder / denser than the air above (inversion layer at about 400 m) (1) • Pollutants cool as they rise and become denser than the warmer air above so they are trapped below the inversion layer and become concentrated (1)	4
06.3	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Cloudless sky allows infrared energy to be radiated from the ground so the ground cools rapidly (1) • The cold ground cools the air in contact with it, producing a cold dense layer near the ground (1) • Low wind velocity means the air layers of different temperatures do not mix (1) • Valley allows cold dense air to collect / drain into the low ground (1)	3
06.4	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Smoke smogs (eg London 1952) (1) • Photochemical smogs (eg Los Angeles, Mexico City) (1) • Build-up of NO_x / carbon monoxide / PM₁₀ in cities (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 241-242; AQA spec 3.4.2.2.	
07.1	<i>1 mark for each point.</i> • Gasification: coal (including deposits too deep to mine) is burnt underground under controlled conditions to produce fuel gases (hydrogen, carbon monoxide, methane) (1) • Allows deep coal to be exploited and gas can be piped / used in gas turbines or for hydrogen with CCS (1) • Liquefaction: coal is converted to liquid hydrocarbons directly with solvents or via gasification and chemical conversion (1) • Provides liquid fuels for vehicles / aircraft that solid coal cannot supply (1)	4
07.2	<i>1 mark for each point.</i> • A solid, ice-like crystalline structure in which methane is trapped within water ice (1) • Found at low temperatures (polar regions / permafrost) or high pressure in ocean sediments around continental margins (1)	2
07.3	<i>1 mark for each point.</i> • Water heating: hot water pumped into sediments melts the hydrate releasing methane (1) • Depressurisation: drilling reduces pressure so methane dissociates from the crystals (1) • CO₂ injection: CO₂ bonds more strongly to the ice and displaces methane (could double as CCS) (1) • Risks: methane not collected escapes to the atmosphere - powerful greenhouse gas / positive feedback; seabed ecology poorly understood; sediment instability / landslides (1)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 187, 190; AQA spec 3.3.4.1.	

Question	Marking guidance	Marks
08.1	1 mark. As the (linear) purity / grade of a mineral deposit decreases, the amount of the mineral present increases exponentially / logarithmically (1)	1
08.2	1 mark for the correct answer. 2400 / 40 = 60 times (1)	1
08.3	1 mark for each point. - Reserves: the amount that can be exploited now, economically, with existing technology (deposits above the cut-off ore grade) (1) - Resource: all the material that could be exploited technically and economically now or in the future (with realistic price rises or new technology), including deep, low-grade or protected deposits (1) - The resource is finite but the reserves change: they rise if prices rise or technology improves, and fall if prices fall (1)	3
08.4	1 mark for each point. - Lowering the cut-off ore grade includes deposits of lower purity in the reserves (1) - Because the amount of material rises exponentially as purity falls, a small drop in COOG adds a very large mass (eg 0.5-1.0% has 4x the copper of 1.0-2.0%) (1)	2
08.5	1 mark for each point. - The total quantity (stock / resource base) is far greater than could ever be used (1) - But most is at purities too low or in locations too inaccessible for current economic exploitation (1) - The challenge is developing technologies (low-energy extraction such as bioleaching, phytomining; better exploration) that make low-grade deposits exploitable; energy needs rise rapidly as ore grade falls (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 135-136; AQA spec 3.2.3.3.	
09.1	1 mark for each point. - pH 5.6 (1) - Atmospheric carbon dioxide dissolves in rainwater to form (weak) carbonic acid (1)	2
09.2	1 mark for each point. - Limestone bedrock (calcium carbonate) is alkaline / basic and neutralises the acids as water percolates through it, so the lake pH stays high (7.2) (1) - Granite does not neutralise acids so lake water in A is acidic (4.6) (1) - Low pH mobilises / dissolves aluminium ions from the soil and rock (0.42 mg dm^{-3} in A) which are toxic to fish (damage gills) (1) - Acids also damage fish eggs / gills (exposed living cells) and dissolve the exoskeletons of invertebrate food; lichens and other food organisms lost (1)	4
09.3	1 mark for each point. - Corrodes metals: railway lines, railings, pipes, pylons, overhead power lines (1) - Dissolves / weakens limestone and marble buildings and statues (eg York Minster) (1)	2

Question	Marking guidance	Marks
09.4	1 mark for each point. - Sulfur dioxide is oxidised (by ozone / other oxidants) to sulfur trioxide (1) - Sulfur trioxide dissolves in water droplets to form sulfuric acid (SO₂ alone gives the weaker sulfurous acid) (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 250-252; AQA spec 3.4.3.2.2.	
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) - Properties of renewables: intermittency, predictability, low energy density, storage, applicability, locational constraints, impacts, technology level, economics - Wind: large offshore resource in shallow seas, falling costs, HAWT / VAWT / direct drive developments; intermittent, needs storage and grid; land-use conflicts - Solar: PV costs falling rapidly, roof space, but low winter insolation in the UK; CSP unsuitable; heat pumps and photothermal for heating - HEP: most large sites used; low-head, helical and micro-hydro extend use; pumped storage vital - Tidal: large range (Severn, Mersey), predictable; barrages costly with high impact, lagoons and in-stream turbines lower impact - Wave: long Atlantic fetch, but technology immature - Biofuels: wastes, anaerobic digestion, energy crops vs food / habitat - Geothermal: limited to low-temperature schemes / heat pumps - Systems issues: storage (pumped HEP, batteries, hydrogen, P2G, V2G), grid management, diversity of sources, energy conservation reducing demand, electrification of transport - Judgement: renewables could supply most electricity with storage and a diverse mix, but heat and transport are harder; transition needs investment and lifestyle change	9
	Model answer - Level 3 (8 marks)	

Question	Marking guidance	Marks
	What renewables can contribute depends on their properties - intermittency, predictability, energy density, storage, applicability, location, impacts, technological maturity and cost - set against the UK's resources. Wind is the largest: shallow seas give a huge offshore resource, costs have fallen and direct-drive and vertical-axis designs keep improving, but it is intermittent, needs storage and grid connections and provokes land-use conflicts onshore. Solar PV costs are falling rapidly and roof space is plentiful, but winter insolation is low and concentrating solar is unsuitable; photothermal panels and heat pumps are better for heating. Most large HEP sites are used, though low-head, helical and micro-hydro extend it and pumped storage is vital for balancing. The Severn and Mersey have large, predictable tidal ranges, but barrages are costly and damaging while lagoons and in-stream turbines have lower impact; the long Atlantic fetch gives wave energy that the technology cannot yet harness reliably. Biofuels from wastes and anaerobic digestion are useful, but energy crops compete with food and habitat, and geothermal is limited to low-temperature schemes.	
	The system matters as much as the sources: storage in pumped HEP, batteries, hydrogen and power-to-gas, vehicle-to-grid, grid management, a diverse mix that smooths supply, and conservation to reduce demand. With these, renewables could supply most of the UK's electricity. Heat and transport are harder, because they need high temperatures or liquid fuels, so electrification and hydrogen would be required. The potential is large, but realising it needs sustained investment and changes in how energy is used.	
	Weaker answer - Level 1 (3 marks)	
	The UK could use lots of renewable energy. It is windy so wind turbines could be built, especially offshore. Solar panels can be put on roofs but it is not very sunny in winter. Tidal power could be used in the Severn estuary because it has big tides. Hydroelectric power can be used in Scotland. Renewables are intermittent so batteries would be needed. Renewables could provide a lot of the UK's energy in the future.	
	What makes the difference	
	• Level 1 mentions the main resources with one fact each; Level 3 assesses each against its properties, constraints and the state of the technology.	
	• Level 3 treats storage, grid and demand as a system and distinguishes electricity from heat and transport - essential for judging 'future energy supplies'.	
	• The conclusion in Level 3 is qualified (most electricity; heat and transport harder; needs investment); Level 1's is unqualified.	
10.2	<i>1 mark.</i> • Eight times (energy proportional to velocity cubed) (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 196-217; AQA spec 3.3.4.1.	

Question	Marking guidance	Marks
11.1	Mineral resources are non-renewable. Discuss how an understanding of geological processes and the development of new technologies can extend the period for which society can rely on mineral resources. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Why non-renewable: formed too slowly; geological concentration processes - hydrothermal, metamorphic, Proterozoic sediments, alluvial, evaporites, secondary enrichment, biological sediments • Understanding processes guides exploration (batholith margins, river gravels, ancient seas) and remote sensing / geophysical techniques • Reserves, resource, stock; Lasky's principle; cut-off ore grade and its drivers • Extraction technologies: mechanisation, bioleaching, phytomining, polymer and bacterial adsorption, seabed nodules • Reducing demand: recycling and its difficulties, Cradle to Cradle, design for disassembly, linear vs circular material flows, dispersal of metals in use • Environmental limits: impacts of mining and their control; energy costs rising with lower grades; conflicts • Judgement: technology can extend supplies substantially but only within energy and environmental constraints; circular economy essential Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 130-147, 373-379; AQA spec 3.2.3, 3.6.3-3.6.4.	25

Question	Marking guidance	Marks
11.2	Discuss how knowledge of the properties of pesticides can be used to reduce their environmental impacts while maintaining food production. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Properties: toxicity (LD50, MDAF, MDNF), specificity, persistence, liposolubility/water solubility, mobility, mode of action (contact/systemic), synergism • Bioaccumulation and biomagnification explained by persistence + liposolubility; DDT and birds of prey as a case study • Comparison of organochlorines, organophosphates, pyrethroids, neonicotinoids with their relative advantages and disadvantages • Reducing impact: restrictions/bans, choice of non-persistent and specific pesticides, systemic pesticides and seed coatings, application timing (night, still days, not in flower), droplet size • Alternatives: cultural control (rotation, barrier/companion crops), biological control, predator habitats, sterile males, pheromone traps, genetic resistance, GM crops, integrated pest management • Trade-offs: yield, cost, human health (organophosphates), resistance evolution, pollinator services; risk:benefit; sustainability and the circular economy • Judgement: no pesticide is perfectly specific; combination of property-based selection and integrated control is most sustainable Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 236-240, 262-266, 314-320; AQA spec 3.4.3.2.8, 3.5.1.1, 3.5.1.5.	25